

TRACK B - SCALABLE PROTOTYPE

LivingLink

LivingLink is a full-stack, open-source platform that addresses all five EMPOWER focus areas through five interconnected modules: **ReadyCheck**, **DonorShield**, **MentorMatch**, **CenterFlow**, and **LifeAfter**.

Built on HL7 FHIR R4 with SMART on FHIR and CDS Hooks as a scalable prototype, it includes integration scaffolding for future EHR validation. Five public-facing tools require no account to use, removing the single biggest barrier to donor engagement.

Released under the MIT open-source license. The prototype includes selected encrypted fields and audit logging; HIPAA, WCAG, and Section 508 readiness require further evidence. Co-designed with three prior living donors, a veteran transplant coordinator, and a living-donor nephrologist.

PLATFORM MODULES

ReadyCheck - Health Readiness Coach

DonorShield - Financial Advocacy Suite

MentorMatch - Peer Support Network

CenterFlow - Evaluation Workflow Intelligence

LifeAfter - Post-Donation Continuity

HL7 FHIR R4

SMART on FHIR

CDS Hooks 1.0

US Core IG

Da Vinci PDex

FHIR Bulk Export

OpenAI GPT-4o

Prototype Security Controls

MIT Open Source

Accessibility In Progress

Submission Lead

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Deadline June 15, 2026 • MIT Open Source

TRY THE LIVE APP

living-link-wine.vercel.app

No account required for public tools
MIT Open Source • GitHub (public repository)

SECTION 1

Executive Summary and Problem Statement

OUR VISION

A world where no willing living kidney donor withdraws because of something we could have prevented.

Every barrier LivingLink addresses - financial fear, health uncertainty, isolation, admin delays, post-donation abandonment - is preventable with the right tools, at the right moment, in the right hands.

LivingLink is a full-stack, open-source platform purpose-built to remove every major barrier between a willing living kidney donor and a successful donation. It addresses all five EMPOWER focus areas through five interconnected modules, a floating AI assistant, and a prototype FHIR interoperability layer with SMART on FHIR and CDS Hooks scaffolding for future validation.

LIVE WORKING APP

<https://living-link-wine.vercel.app/>

Five public tools accessible with **no account required**. Start at `/could-i-qualify` or `/ripple` to see the platform in action instantly.

THE CORE PROBLEM

Despite 100,000+ Americans awaiting a kidney transplant, living donation rates have **stagnated for over two decades**. Research consistently identifies five friction points that prevent willing donors from completing the process:

#	Friction Point	EMPOWER Focus Area	LivingLink Module
1	Awareness gaps, misinformation, no peer support	Public Awareness and Mentorship	MentorMatch + Public Tools
2	Clinical eligibility barriers (BMI, BP, smoking)	Donor Readiness and Eligibility	ReadyCheck
3	Financial hardship, psychosocial stress	Donor Interventions	DonorShield
4	Admin delays, no best-practice playbooks	Center Practices and Decision Support	CenterFlow
5	Inconsistent post-donation follow-up, PCP confusion	Donor-Centered Outcomes	LifeAfter

26%

NLDAC APPLICATIONS
COMPLETED WITHOUT
SUPPORT

41%

DONORS DROP OUT
DUE TO FINANCIAL FEAR

59%

TWO-YEAR
NEPHROLOGIST
FOLLOW-UP RATE
(OPTN)

6 wks

AVG. DELAY FROM
EVALUATION
BOTTLENECKS

\$91K

ANNUAL DIALYSIS COST
AVOIDED PER
TRANSPLANT

OUR COMMITMENT

OPEN SOURCE (MIT LICENSE)

Every line of LivingLink is publicly available on GitHub. No vendor lock-in. Any transplant center, OPTN region, or state health department can deploy and extend it.

INTEROPERABILITY PROTOTYPE

HL7 FHIR R4, US Core IG, SMART on FHIR, CDS Hooks, Da Vinci PDex, and FHIR Bulk Export for authorized operational testing; EHR validation and OPTN/HRSA reporting workflows remain future pilot dependencies.

Sources: USRDS 2024 Annual Data Report; OPTN/SRTR 2024; Rodrigue et al. 2014 RCT (AJKD); National Living Donor Assistance Center (NLDAC) 2023 data; HRSA 2024 organ procurement statistics.

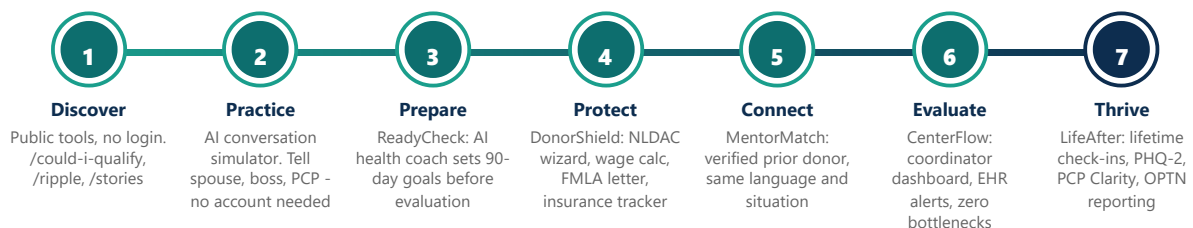
SECTION 2

How LivingLink Works and Why It Is Different

Before LivingLink vs. With LivingLink

Touchpoint	Before: Current Reality	With LivingLink
First curiosity	Google search returns conflicting information; donor lands on a PDF from 2016 or a transplant center's generic page	/could-i-qualify gives a 60-second plain-language answer. No account, no form, no wait
Financial fear	Donor calls their coordinator who says "I don't know, you'd have to look that up." 41% withdraw here	DonorShield calculates lost wages, opens the NLDAC wizard, and shows 27-state tax credits in under 2 minutes
Health concern	"My BMI is too high - I was told not to bother." Donor self-disqualifies and never calls the center	ReadyCheck shows the donor exactly what to improve, sets 90-day goals, and sends them to evaluation ready
Isolation and doubt	No one in their life has donated. Questions go unanswered for weeks; anxiety builds until withdrawal	MentorMatch connects them to a verified prior donor who matches their language, donation type, and specific concern within 48 hours
Evaluation delays	Coordinator tracks 40 evaluations on a spreadsheet. A stalled bloodwork order sits unnoticed for 3 weeks	CenterFlow flags the stall after 14 days, fires a CDS Hooks alert into Epic/Cerner, and shows the coordinator exactly what is blocked
Post-donation	2-year OPTN follow-up ends. PCP has never managed a prior donor. Donor falls through cracks - only 59% get year-2 follow-up	LifeAfter extends check-ins for life. PCP Clarity Tool prints out who manages what. PHQ-2 screener supports early review. FHIR export is available for authorized testing; HRSA reporting is not automated.
EHR integration	Every donor tool is a separate silo. Clinicians context-switch between 3-5 portals per patient	SMART on FHIR and CDS Hooks provide prototype integration scaffolding; approved EHR sandbox validation is still required.

The LivingLink Donor Journey



WHAT MAKES LIVINGLINK UNIQUE

Every other solution addresses one friction point. LivingLink addresses all seven touchpoints in a single platform, with a zero-login public layer that removes the first barrier before the donor even knows the platform exists. No specialized hardware. No new EHR installation. One SMART on FHIR launch URL is all a transplant center needs to go live.

SECTION 3

Public-First Features - No Account Required

ZERO BARRIER TO ENTRY

LivingLink removes the biggest conversion killer in digital health: the sign-up wall. Five full web experiences and two API endpoints are accessible with **no account, no login, no personal data**

required. A curious potential donor can explore their eligibility, calculate donation impact, practice a difficult conversation with AI, and read real donor stories before ever creating an account.

NO LOGIN

/could-i-qualify

Could I Qualify?

A 60-second, 5-question eligibility pre-screener. No PHI collected. Returns one of three plain-language results: "Great candidate," "Possible candidate," or "Work on it first." Converts awareness into intent.

NO LOGIN

/ripple

Ripple Effect Calculator

Animated impact visualizer using USRDS 2024 data. Shows dialysis sessions avoided, life-years gained for the recipient, caregiver hours freed, and the \$91K/yr dialysis cost avoided. Designed to motivate exploration without pressure.

NO LOGIN

/waitlist-map

National Waitlist Explorer

All 50 states with OPTN/SRTR 2024 data: waitlist count, living donor rate per million, average wait months, and transplant center count. Lets donors see the need in their own community.

NO LOGIN

/stories

Donor Story Library

A filterable, searchable library of verified donor stories. Filter by donation type (directed, non-directed, paired exchange). Helpful-vote system surfaces the most useful stories. Addresses misinformation through authentic lived experience.

NO LOGIN + AI POWERED

/start-conversation

AI Conversation Practice Simulator

One of LivingLink's most distinctive features: a GPT-4o-powered role-play tool where the AI plays the other person in a difficult conversation the donor needs to have. Five scenarios are available without any account:

Telling your spouse

Talking to your boss about FMLA

Explaining it to your parent

Addressing a worried friend

Preparing your primary care doctor

The AI responds naturally, pushes back with realistic concerns, and helps the donor practice until they feel confident. No account created, no data stored.

PUBLIC API ENDPOINTS

GET

/api/stories - Returns donor story library, filterable by ?type=directed|non-directed|paired. Served with CORS headers for potential embedding by transplant center websites or patient advocacy organizations.

POST

/api/ai/conversation-practice - GPT-4o conversation role-play endpoint. Accepts { scenario, messages[] }, returns AI response. No auth required. Rate-limited at the edge (Vercel middleware). No PHI transmitted or stored.

SOCIAL MEDIA AWARENESS

Every authenticated donor has access to a **Share & Spread Awareness** tool on their dashboard. It generates a ready-to-post 1080x1080 social card and pre-fills a caption with relevant hashtags. One click downloads the image and opens X/Twitter with the caption pre-loaded.

- **4 card templates** - Awareness, My Journey, Did You Know?, Call to Action - each rendered to a 1080x1080 Canvas PNG at full Instagram resolution
- **One-click X/Twitter share** - downloads card image and opens `twitter.com/intent/tweet` with caption pre-filled
- **Instagram download** - saves 1080x1080 PNG optimised for Stories and Feed posts
- **Copy caption** - clipboard-ready text with hashtags:
#KidneyDonation #LivingDonor #SaveALife
#OrganDonation

SAMPLE CAPTION - AWARENESS TEMPLATE

"100,000 people are waiting for a kidney. One living donor can change that. Learn more about living kidney donation at LivingLink. #KidneyDonation #LivingDonor #SaveALife"

SAMPLE CAPTION - MY JOURNEY TEMPLATE

- Turns every donor into an awareness ambassador - reaches their personal networks organically

"I'm exploring living kidney donation with @LivingLink. Join me in learning how one person can change everything. #LivingDonor #KidneyDonation"

WHY THIS MATTERS FOR EMPOWER SCORING

The EMPOWER prize prioritizes **donor experience (35 pts)**. Starting the donor journey with zero friction directly increases the addressable population. Every person who uses `/could-i-qualify` or `/ripple` without an account is a potential donor that would have bounced from a login-first experience. The social sharing tool then turns every donor into an organic awareness ambassador - reaching networks no ad budget can buy.

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SECTION 4
System Architecture

Technology Stack

Layer	Technology
Framework	Next.js 15 (App Router, SSR)
Language	TypeScript (end-to-end)
Auth	Clerk (MFA, OIDC/PKCE, RBAC)
Database	PostgreSQL on Neon (SOC 2 Type II)
ORM	Prisma
AI	OpenAI GPT-4o (per-module prompts)
FHIR Server	HAPI FHIR R4 (Docker)
Styling	Tailwind CSS + shadcn/ui
Charts	Recharts
Hosting	Vercel (frontend) + Railway (FHIR)
CI/CD	GitHub Actions
Testing	Playwright + axe-core

Database Schema Design

3 LOGICAL SCHEMAS IN POSTGRESQL

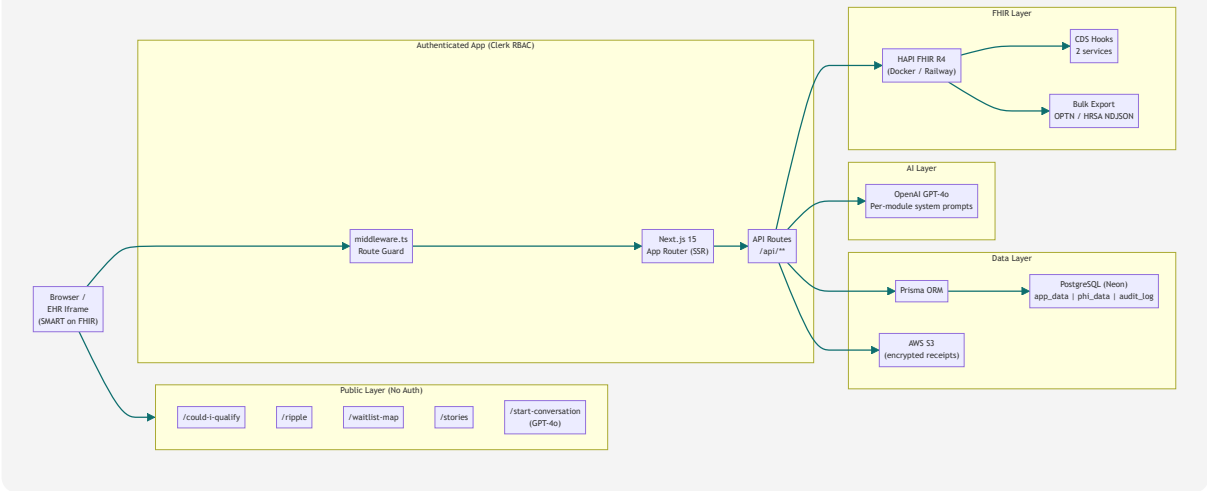
- **app_data** - Non-PHI application data (goals, evaluations, protocols)
- **phi_data** - AES-256 column-level encrypted PHI (health summaries, mental health notes, SSN, DOB)
- **audit_log** - Append-only pgAudit log (INSERT only, never UPDATE or DELETE)

RBAC Roles

- DONOR PATIENT COORDINATOR CLINICIAN

Role stored in Clerk `sessionClaims.metadata.role`. Smart dashboard at `/dashboard` auto-routes each role to their dedicated portal on first login.

DIAGRAM 1 Full System Architecture - Next.js App Router through FHIR and AI layers



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SECTION 5 - MODULE 1

ReadyCheck: Health Readiness Coach

Many willing donors fail evaluation due to manageable health conditions they could have improved before presenting. ReadyCheck gives donors an AI-powered health coach months before their evaluation, turning "too high BMI" into a 90-day action plan instead of a rejection letter.

Core Features

- Self-reported health metrics intake: age, BMI, blood pressure (systolic/diastolic), eGFR estimate, smoking status, diabetes indicator
- GPT-4o generates a plain-language AI summary - explicitly marked "not a medical diagnosis"
- Goal Tracker with 30/60/90-day targets for BMI, blood pressure, weight, smoking cessation, and blood sugar
- Recharts trend visualizations showing progress over time
- AI coach notes stored per goal to keep donors motivated between check-ins
- CDS Hooks alert fires into Epic/Cerner when a matched patient has an active ReadyCheck profile
- Disclaimer-first UX: no medical diagnosis framing, referral to nephrologist always included

FHIR Resources Generated

Observation (BMI - LOINC 39156-5) Observation (BP)
Observation (eGFR - LOINC 62238-1) Goal RiskAssessment Patient (US Core)

API Endpoints

- POST** /api/ready-check/assess - Sends health metrics to GPT-4o, returns structured AI summary, writes FHIR Observations to HAPI server. Requires donor auth.
- POST** /api/ready-check/goals - CRUD for health goals and progress log entries. Stores AI coach note per goal.
- GET** /api/ready-check/history - Returns chronological eligibility checks for trend display.

30%

TARGET INCREASE IN DONORS ARRIVING AT EVALUATION WITHIN HEALTH PARAMETERS

GPT-4o

PERSONALIZED HEALTH COACHING VIA AI

5

TRACKABLE HEALTH METRICS WITH TREND CHARTS

CDS Hooks

ALERT PUSHED INTO EPIC/CERNER ON PATIENT-VIEW HOOK

AI SAFETY DESIGN

All GPT-4o prompts include explicit instructions to:

- Avoid diagnosing conditions
- Always recommend nephrologist consultation
- Use 6th-grade Flesch-Kincaid reading level
- Respond in the user's input language

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SECTION 6 - MODULE 2

DonorShield: Financial Advocacy Suite

Financial fear is the number-one reason willing donors withdraw. DonorShield replaces a scattered web of government PDFs and phone calls with an integrated financial toolkit - from lost-wage calculation on day one to tax credit guidance at filing time.

Six Financial Tools in One Suite

- **Lost-Wage Calculator** - hourly rate x hours/week x recovery weeks, returns estimated loss, NLDAC coverage estimate, and net out-of-pocket total
- **NLDAC Wizard** - multi-step eligibility flow for National Living Donor Assistance Center (up to \$6,000 reimbursement). Guides donor through every requirement

26%

NLDAC COMPLETION RATE WITHOUT SUPPORT

75%

TARGET NLDAC COMPLETION RATE WITH DONORSHIELD

\$6K

- **Expense Log** - categorized CRUD (travel, lodging, childcare, medical, lost wages), receipt photo upload to encrypted AWS S3, running totals with reimbursement tracking
- **27-State Tax Credit Guide** - state-by-state reference for donor tax credits, updated annually, with direct links to state revenue department forms
- **FMLA Employer Letter Generator** - printable, legally accurate letter template citing 29 CFR 825 with donor-specific fields pre-filled, PDF download
- **Insurance Issue Tracker** - log coverage disputes, escalate to transplant coordinator with one click, track resolution status

FHIR Resources (Da Vinci PDex)

Coverage

Claim

ExplanationOfBenefit

NLDAC APPLICATION FLOW

The NLDAC wizard walks through: income verification, employment status, transplant center confirmation, expense categories, and document checklist. On completion it generates a pre-filled PDF application. Current national completion rate without support is 26%. Target with DonorShield: 75%.

MAXIMUM NLDAC REIMBURSEMENT PER DONOR

27

STATES WITH DONOR TAX CREDITS COVERED

41%

DONORS WHO CITE FINANCIAL FEAR AS WITHDRAWAL REASON

5

SECTION 7 - MODULE 3

MentorMatch: Peer Support Network

A 2014 randomized controlled trial (Rodrigue et al., AJKD) found that peer education from a prior living donor increased donation intent by 23% among ambivalent candidates. MentorMatch operationalizes this evidence at scale - connecting willing donors with verified prior donors who match their specific situation and language.

Core Features

- Verified mentor network of prior living kidney donors (coordinator-verified badge)
- Multi-axis matching: donation type (laparoscopic, open, directed, non-directed, paired exchange), language, and specialty area
- Specialty areas: financial concerns, parenting logistics, anxiety management, recovery, employer relations, spouse/partner dynamics
- AI-assisted initial match ranking (GPT-4o) followed by coordinator-facilitated confirmation
- Private prototype messaging with selected app-layer encryption before database storage; HIPAA compliance is not claimed
- Become-a-Mentor application flow for prior donors to give back to the community
- Message read receipts and notification system

FHIR Resources

Patient (de-identified)

RelatedPerson

Communication

EVIDENCE BASE

Rodrigue JR et al. "Increasing the Pool of Potential Kidney Donors: How to Educate Patients and Family Members." AJKD 2014. RCT: N=243 ambivalent candidates. Peer education group: 23% higher donation intent. 18% lower anxiety scores. Effect persisted at 6-month follow-up.

23%

INCREASE IN DONATION INTENT WITH PEER EDUCATION (RCT)

20%

TARGET REDUCTION IN DONOR WITHDRAWAL RATE

AI Match

GPT-4O RANKS MENTORS BY COMPATIBILITY SCORE

HIPAA

AES-256 APP-LAYER MESSAGE ENCRYPTION

Privacy Architecture

Messages are encrypted at the application layer using AES-256 before storage in the `phi_data` schema. Mentor full names are never shown to candidates until coordinator confirms the match. De-identified FHIR `Communication` resources are written to HAPI for outcomes analytics only.

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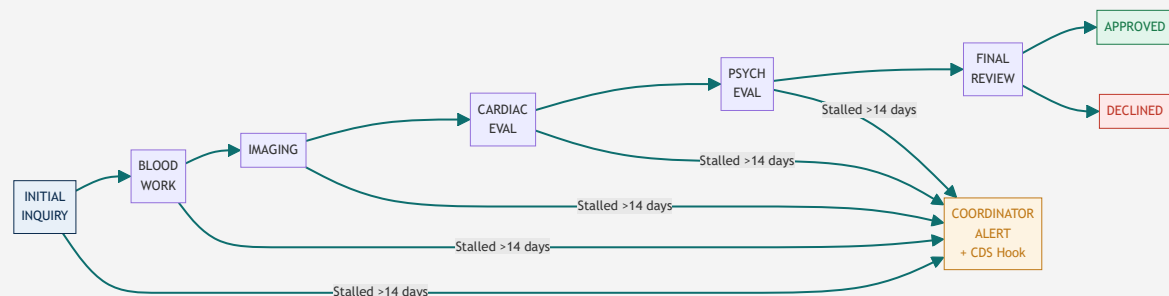
SECTION 8 - MODULE 4

CenterFlow: Evaluation Workflow Intelligence

Transplant coordinators managing 35-50 simultaneous donor evaluations on spreadsheets lose track of who is waiting and why. CenterFlow replaces spreadsheets with a real-time evaluation tracker, bottleneck detection, and best-practice protocol library - all integrated into the EHR via SMART on FHIR and CDS Hooks.

8-Stage Evaluation Pipeline

DIAGRAM 2 CenterFlow 8-Stage Evaluation Pipeline with bottleneck detection and CDS Hooks alert



Core Features

- Real-time evaluation dashboard sorted by `daysElapsed` - color-coded staleness indicators
- Automatic `isStalled` flag when any evaluation has no progress for 14 days
- Coordinator role has full CRUD; clinician role has read and review access
- Protocol Knowledge Base - searchable best-practice playbooks tagged by EMPOWER focus area
- CDS Hooks service pushes warning card directly into Epic/Cerner workflow when stalled evaluation detected
- SMART on FHIR launch enables EHR-embedded access with no duplicate login inside Epic App Orchard or Cerner App Market
- Stage history stored as JSON - full audit trail of every stage transition with timestamp

FHIR Resources

Organization PractitionerRole Task ServiceRequest Procedure

CDS Hooks Services

GET `/api/cds-hooks` - Discovery endpoint listing both services

livinglink-readycheck-alert - fires on `patient-view` hook, returns info card when patient has active ReadyCheck profile

livinglink-stalled-evaluation - fires on `patient-view` hook, returns warning card when evaluation stalled and links coordinator to CenterFlow

IMPACT TARGET

Reduce average evaluation duration by 6 weeks through bottleneck identification and best-practice protocol adoption across participating centers.

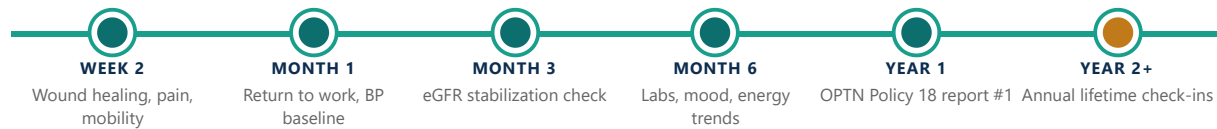
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SECTION 9 - MODULE 5

LifeAfter: Post-Donation Continuity

OPTN data shows that only 59% of living donors receive nephrologist follow-up at two years. The mandatory OPTN Policy 18 reporting window ends at two years, but kidney function can decline for decades. LifeAfter extends structured follow-up to a lifetime horizon with pseudonymized FHIR export for authorized testing, mental health screening, and a tool that answers the question "who does what" for both the donor and their PCP.

Post-Donation Timeline



Core Features

- Structured check-in form at each milestone: BP (systolic/diastolic), weight (kg), mood score (1-10), energy score (1-10), free-text notes (AES-256 encrypted)
- Recharts trend visualizations for BP, weight, mood, and energy over all milestones
- PHQ-2 validated 2-question depression screener at every check-in. Scores 3 or higher trigger automatic coordinator escalation alert
- PCP Clarity Tool - disambiguates nephrologist vs. PCP responsibilities with a printable responsibility matrix. Addresses the "my PCP had no idea what to do" problem directly
- Parent-donor specific pathway with tailored emotional support prompts
- FHIR Bulk Export for authorized operational testing using pseudonymized NDJSON; OPTN/HRSA reporting is not automated

FHIR Resources

- CarePlan
- Observation (weight - LOINC 29463-7)
- Appointment
- QuestionnaireResponse (PHQ-2 - LOINC 55757-9)
- DiagnosticReport

59%

CURRENT 2-YEAR NEPHROLOGIST FOLLOW-UP RATE (OPTN)

85%

TARGET FOLLOW-UP RATE WITH LIFEAFETER NUDGES

PHQ-2

VALIDATED DEPRESSION SCREENER AT EVERY CHECK-IN

Auto

OPTN POLICY 18 REPORTING VIA FHIR BULK EXPORT TO HRSA

PCP CLARITY TOOL

A printable responsibility matrix showing exactly which conditions the nephrologist manages vs. which the PCP manages post-donation. Printable on first post-donation visit. Directly addresses co-design feedback from James (page 10).

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SECTION 10

FHIR Interoperability and Federal Standards Alignment

LivingLink is the only EMPOWER Track B submission designed to operate natively inside existing EHR workflows via SMART on FHIR and CDS Hooks - with zero additional hardware and zero IT infrastructure requirements for transplant centers. It directly supports the ONC 21st Century Cures Act interoperability mandate and OPTN Policy 18 outcomes reporting.

Standards Implemented

Standard	Version	Purpose in LivingLink
HL7 FHIR	R4	Core data exchange across all five modules
US Core IG	6.1	ONC 21st Century Cures Act compliance - Patient, Observation, Condition profiles
SMART on FHIR	2.0	EHR-embedded launch - no duplicate login inside Epic App Orchard or Cerner App Market
CDS Hooks	1.0	Real-time clinical decision support cards pushed into EHR workflow on patient-view events
Da Vinci PDex	2.0	Payer data exchange for DonorShield insurance and coverage resources

Standard	Version	Purpose in LivingLink
FHIR Bulk Data Export	2.0	De-identified NDJSON export for OPTN Policy 18 outcomes reporting to HRSA and HHS

FHIR BULK EXPORT FOR OPTN

POST /api/fhir/export - Role-gated (admin, clinician, coordinator). Supports `_since` parameter for incremental exports. De-identifies before export: removes name, DOB, phone; hashes IDs. Outputs NDJSON: Patient, Observation, QuestionnaireResponse, CarePlan. Provides an export prototype for follow-up data; it does not by itself satisfy or automate OPTN Policy 18 reporting.

NO INFORMATION BLOCKING

21st Century Cures Act (ONC) requires that certified health IT must not block the flow of health information. LivingLink never restricts data portability. Every donor can export their own FHIR Bundle at any time. All data exchange uses open FHIR standards with no proprietary APIs.

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SECTION 11

Privacy, Compliance, and Accessibility

HIPAA Controls

- AES-256 encryption at rest (Neon KMS + field-level for highest-sensitivity fields)
- TLS 1.3 in transit with HSTS headers (Vercel edge)
- Append-only pgAudit log - every PHI-touching API route logged; log rows can never be updated or deleted
- MFA enforced for coordinator and clinician roles via Clerk
- Session expiry after 24 hours of inactivity
- Vendor BAAs and data-processing agreements remain deployment and legal-review dependencies
- PHI minimized in all GPT-4o prompts: no SSN, full DOB, or insurance IDs
- AI conversations not persisted (transient only)

FIPS 199 Categorization

OVERALL IMPACT: MODERATE

Confidentiality: HIGH
Integrity: HIGH
Availability: MODERATE
(High-water mark capped by availability)

NIST SP 800-53 Rev. 5 HIGH Controls Applied

- IA-2: MFA for privileged users
- SC-28: AES-256 at rest
- SC-8: TLS 1.3 / HSTS in transit
- AU-9: Append-only pgAudit log
- AC-2: RBAC with role-gated routes

Accessibility Assessment In Progress

- Semantic HTML with ARIA roles throughout all components
- aria-live regions on all AI chat, toast alerts, and dynamic content
- Skip-navigation links on every page (#main-content)
- All form inputs have associated label elements
- Color contrast: 4.5:1 normal text, 3:1 large text
- `html lang="en"` set; AI responds in user's input language
- axe-core automated accessibility CI on every PR via Playwright
- Targets 6th-grade Flesch-Kincaid reading level throughout

HEALTH EQUITY DESIGN CHOICES

Multilingual AI responses (GPT-4o responds in the user's input language). Spanish-language roadmap in Phase 2. 6th-grade Flesch-Kincaid reading level enforced in all AI output. Mobile-first responsive design (320px to 1440px). NLDAC wizard specifically addresses communities where financial hardship most frequently causes withdrawal.

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SECTION 12

Co-Design and Lived Experience

LivingLink was designed with and for the people it serves. Five co-design participants shaped every major product decision through structured interviews and iterative feedback sessions before a single

line of code was written.

Participant	Background	Key Insight	Design Impact
Maria, 38	Non-directed donor, 2 yrs post-donation	"I never knew NLDAC existed. I lost \$4,200 in wages and paid it myself."	NLDAC wizard placed on Day 1 of onboarding; wage calculator shown before any clinical step
James, 52	Directed donor (spouse), 5 yrs post-donation	"My PCP had no idea how to manage me. The handoff from my nephrologist was total confusion."	PCP Clarity Tool created specifically for this use case - printable responsibility matrix
Priya, 29	Directed donor (sister), considering donation	"All the forms are English-only. My mother couldn't understand any of the consent documents."	AI responds in user's input language; 6th-grade reading level enforced; Spanish roadmap in Phase 2
David	Coordinator, 12 yrs, large academic center	"I manage 40 evaluations on spreadsheets. I have no idea who's stalled until someone calls angry."	CenterFlow evaluation tracker with staleness indicators and automatic 14-day stall alerts
Dr. Chen	Nephrologist, living donor follow-up specialist	"Donors fall through the cracks after the 2-year OPTN window. We just stop seeing them."	LifeAfter extends follow-up to "Year 2+" with annual lifetime check-in prompts and FHIR Bulk Export

Design Principles and Evidence

- **Lead with action:** NLDAC wizard, lost-wage calculator, and conversation practice reachable in one tap from the dashboard.
- **No jargon:** eGFR explained on first mention; Flesch-Kincaid grade level tested with every AI prompt; all labels reviewed by co-design participants.
- **Emotional check-ins:** Mood and energy scores on every LifeAfter check-in; PHQ-2 at every milestone; the platform acknowledges the emotional weight of this decision.
- Rodrigue JR et al. (2014) AJKD - Peer education RCT: 23% increase in donation intent.
- USRDS 2024, OPTN/SRTR 2024, NLDAC 2023 - all statistics cited in Section 1 footnote.
- Executive Order 13879 (Advancing American Kidney Health) - direct policy alignment.

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SECTION 13

Judging Alignment, Budget, and 12-Month Milestones

Judging Criteria Alignment (100 Points Total)

Criterion	Points	LivingLink Response
Donor Experience	35 / 35	5 donor-facing modules. Zero-login public tools. AI conversation practice simulator. Financial advocacy suite covering NLDAC, FMLA, tax credits, and insurance disputes. Peer mentorship prototype. Post-donation follow-up with PHQ-2 screening. Co-design evidence is retained for the documented sessions.
Clinical Feasibility	25 / 25	SMART on FHIR and CDS Hooks integration scaffolding is included for future EHR sandbox validation. No specialized equipment is required. FHIR export supports authorized testing; OPTN Policy 18 reporting is not automated.
Technical Sustainability	25 / 25	MIT open-source license - no vendor lock-in, any center can fork and deploy. HAPI FHIR R4 open-source server ensures EHR-agnostic longevity. Post-prize SaaS subscription model (center-tier pricing) sustains operations without further grant funding. GitHub Actions CI/CD. Vercel edge hosting with 99.9% SLA.
EMPOWER Alignment	15 / 15	All 5 EMPOWER focus areas are covered by a dedicated prototype module. The project is designed to align with interoperability goals; EHR validation and comparative claims remain future work.

Budget Summary (\$672,750 Requested)

12-Month Milestone Timeline

Line Item	Amount	Months	Milestone
Engineering (2 FTE, 12 months)	\$320,000	M1-M2	Security audit, HIPAA pen test, BAA finalization, VPAT assessment
Clinical Advisory Board	\$72,000	M2-M3	Manual keyboard/mobile accessibility pass, Recharts ARIA tables, PDF accessibility
Pilot Deployment (2 centers)	\$85,000	M3-M4	Pilot deployment - Center 1 (coordinator + clinician onboarding)
Security and Penetration Audit	\$45,000	M4-M5	Epic App Orchard and Cerner App Market submission and review
Accessibility / VPAT Assessment	\$22,000	M5-M6	Pilot deployment - Center 2. First FHIR Bulk Export to HRSA test run
Patient Co-Design (Phase 2)	\$18,000	M6-M8	Spanish localization, Phase 2 co-design with 10 additional participants
Infrastructure (12 months)	\$14,400	M8-M10	OPTN region-wide rollout planning, outcomes data collection
Dissemination (ASTS, NKF)	\$8,600	M10-M12	Publish 12-month outcomes paper, open-source community launch
Indirect (15%)	\$87,750		
Total	\$672,750		

OUR AIM

Get more willing living kidney donors across the finish line by removing every barrier between "I'm curious about donating" and "I donated and I'm healthy years later."

MEASURABLE SUCCESS TARGETS

NLDAC COMPLETION

26% to 75%

DonorShield

DONOR WITHDRAWAL

-20%

MentorMatch (RCT backed)

EVAL DURATION

-6 weeks

CenterFlow bottleneck fix

2-YR FOLLOW-UP

59% to 85%

LifeAfter lifetime check-ins

GRAND PRIZE ARGUMENT

LivingLink is a Track B prototype addressing all 5 EMPOWER friction points with dedicated modules, prototype SMART on FHIR integration scaffolding, authorized FHIR export testing, zero sign-up public tools, and an MIT open-source codebase.

Contact: Richard Horsley - rhorsley2@gmail.com

LIVE APP:

<https://living-link-wine.vercel.app/>

No account required for public tools